ONE PAGE SUMMARY

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CUSTOMIZATIONS AND WORKAROUNDS DURING THE IMPLEMENTATION OF AN OPEN-SOURCE ELECTRONIC MEDICAL RECORD SYSTEM

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Introduction

The use of technology in healthcare is associated with many advantages such as cutting down healthcare costs and improving the quality of care. It is estimated that, by improving health care efficiency and safety, the widespread adoption of electronic medical record (EMR) systems in the US could save more than $81 billion annually (Hillestad et al., 2005). Despite the importance of EMR systems, adoption in North America has been hampered by many issues. Only 4% of physicians in the US reported having a fully functional EMR system and 13% have reported having a basic system (DesRoches et al., 2008). The situation in hospitals is even worse. A recent study reported that only 1.5% of US hospitals have a comprehensive EMR system and only 7.6% have a basic system (Jha et al., 2009).

While cost is identified as the main barrier to adopting an EMR (DesRoches et al., 2008; Jha et al., 2009), recent studies looked at other social barriers such as the complexity of the implementation and the incompatibility between the long established traditions of medical work and the new technology (Safadi et al., 2010a). Similar results are reported by other research. Kaissi et al. (2007) found that the fit between organizational culture—beliefs, values and artifacts—and organizational structure—recurrent patterns of routine interaction and behavior—affected the rate of medication errors when using EMR and drug interaction systems.

Previous research (Safadi et al., 2010b) imply that for EMR systems to succeed, they should be highly flexible and allow for customizations and modifications in order to match the particularities of the medical practices in the deploying site. This requirement may not be easily satisfied by commercial systems at reasonable cost. Commercial vendors aim to maximize their revenues and minimize the development cost by offering general software packages—software for the masses. This trade-off between customization and cost is a major issue that inhibits the adoption of medical software.

Does Open-Source EMR Provide a Solution?

The open-source software development movement has gained momentum in the last years and emerged as a serious competitor to the traditional proprietary software development model. Several open-source EMR systems were developed in North America. A comprehensive review of such systems is provided in Faus et al. (2008). The success of open-source software may translate to the medical field. More specifically, the unique features of open-source software may alleviate some of the obstacles

traditionally associated with proprietary systems. Advantages of the open-source solution include: the low cost (no software cost, no licensing cost) of acquisition and maintenance, greater possibility of customization, lower exposure to vendor failure or product termination. On the other hand, several unique disadvantages may hurt the open-source EMR: lack of integration with existing vendor-based hospital systems, a fragmented development effort, a limited number of firms that support installation and training, and a lack of clear software development roadmap.

An open-source EMR differs from typical open-source software in the amount of customization required to make it workable in a specific context. The EMR has to comply with the legal requirements, drug nomenclatures and references, billing systems, integration with state agencies, and local practices. Given the high amount of customization the involvement of users (i.e. medical practitioners) in the development process is crucial for its success. While the recent software engineering literature stresses the importance of feedback in the software development process (Sommerville, 2000), it limits the scope of feedback to explicit communication between developers and users of the system.

The previous discussion suggests that the implementation of an open-source EMR may differ from that of a commercial EMR. In several domains, successful open-source products emerged and competed with proprietary products. For example, the Linux operating system, the Apache web server, and the Mozilla Firefox browser are successful open-source products that gained market success over their proprietary competitors. Yet, it is unclear whether the open-source success extends to the medical domain. In addition, the dynamics of an open-source EMR may differ significantly from that of a commercial system. In particular, the implementation of an open-source EMR requires more involvement and participation from medical practitioners in order to customize and operate the system. Does open-source EMR provide a solution for the challenges surrounding the implementation of EMR? In order to answer this question, this paper proposes to study the implication of open-source EMR systems on clinical and administrative practices in ambulatory healthcare.

Literature Review

The research topic proposed in this paper draws on literature from multiple disciplines. The implementation of EMRs is not merely a technical problem of designing and implementing the appropriate interface. The medical record is deeply interwoven within the structure of medical work, and different medical record systems represent different notions of how medical work is organized (Berg et al., 1997). This section starts by reviewing the literature on the impact of EMRs. Then it moves to discuss open-source EMR. Finally, it synthesizes the discussion and identifies an opportunity for contribution.

Impact of EMR on Healthcare

It is agreed upon that electronic medical record systems bring major benefits to the healthcare sector, mainly by reducing the cost of healthcare. A study by Hillestad et al. (2005) estimates that, by improving health care efficiency and safety, the widespread adoption of EMR systems in the US can save more than $81 billion annually. This figure could double if advanced functionalities of EMR systems such as enabled prevention and management of management of chronic disease are realized. Walker et al. (2005) report similar findings—the value of fully standardized health care information exchange and interoperability among providers and consumers could be more than $77.8 billion per year. Non-standardized implementations yield smaller positive returns as well.

Impact of EMR on Medical Practice

In an ethnographic analysis, Ventres et al.(2006) found that the introduction of an EMR intro practice triggered several unexpected cognitive and social changes in the clinical encounter. The study identified 14 factors of how EMRs are used and perceived. The factors are classified into four classes:

1. Spatial: impact of the location of the EMR on physician-patient interaction.
2. Relational: how physicians and patients perceptions of the EMR affect its use.
3. Educational: issues of physicians training and patients understanding of the EMR.

4. Structural: institutional and technological factors that affect the use of the EMR.

Taking the relational impact, Garrido, Jamieson, Zhou, Wiesenthal, and Liang (2005) studied the impact of EMR systems on the number of visits in ambulatory care clinics. The study found that an EMR system helped in reducing the number of visits that are unnecessary or marginally productive. Specifically, two years after the implementation of EMRs in different sites in the US, age adjusted primary care visit fell by 11% and specialty care visits dropped by 6%. Combined together age adjusted rates of office visits decreased by 9%.

Tamblyn et al. (2006) studied the implementation of an integrated electronic prescribing and drug management system in primary care clinics. They found that physicians believed that the system would improve the continuity of care, and that they are more willing to adopt it in more complex, fragmented situations.

Several studies looked into the cultural issues surrounding the implementation of EMR systems. Kaissi, Kralewski, Dowd, and Heaton (2007) found that the fit between organizational culture—beliefs, values and artifacts—and organizational structure—recurrent patterns of routine interaction and behavior—affected the rate of medication errors when using electronic medical record and drug interaction systems. They conclude that administrators should consider the culture of their organizations before implementing new structures such as EMRs.

Supporting this finding, Baron et al. (2005) highlighted several key decisions they made when implementing an EMR system in their clinic. One key decision was to minimize the IS impact on physician-patient interaction. To achieve this goal, tablet computers with wireless connection were chosen. Another key decision was not to impose any additional requirement on the physicians because of the new system.

When the system implementation contradicts with cultural values or induces undesirable changes in the work process, episodes of resistance occur. Rivard, Lapointe, and Kappos (2009) show that difficulty of implementing an EMR system depends on its effects on the different cultural values in the hospital. They identified four values: quality of care, efficiency of clinical practices, medical dominance, and professional autonomy. Then they classified these values using Martin's three dimensions of a culture: integration, differentiation, and fragmentation (Martin, 1992). Quality of care and clinical practices are integrative values—their interpretation is shared across all medical workers. Medical dominance and professional autonomy are differentiating values—they differ depending on the subgroup (doctors, nurses, and technicians). Implementation is facilitated when it is consistent with the four values.

Differentiating values are closely related to the work process and power. Other studies support this finding. Beaudry and Pinsonneault (2005) present a framework that relies on the notion of coping to explain users' adoption of an IT system. Once the system is implemented, users try to relate the systems to their own objectives and benefits. Users assess the advantages and opportunities that they get from IT and at the same time evaluate the threats and risks that the system poses. Users seek to increase their efficiency and effectiveness, minimize the negative consequences of IT, and restore their emotional stability. Lapointe and Rivard (2005) studied the implementation of three EMR systems in three hospitals. In one case, they found that physicians' resistance to the system was triggered by its slowing down their process of carrying out medical procedures. Physicians' resistance was passive at the beginning but started to become active when the distribution of power between physicians and nurses was affected—physicians felt the system is making them do more clerical tasks.

In summary, the literature on the impact of EMR suggests that the introduction of EMR in the clinic leads to unexpected changes. Although research has indicated many benefits of using EMR in term of reducing medical errors, it is important recognize that cultural changes may not be welcomed in an old profession such as medicine. As stated previously, for EMR to succeed it has to fit the medical practices and satisfy medical practitioners, especially doctors. This research will look on how open-source EMR manage to fulfill these two requirements.

Open-Source EMR

Open-source was traditionally used by tech savvy users. This trend is however changing and we are seeing more open-source software replacing proprietary software in many areas (Muffatto, 2006). Several open-source EMR systems were developed in North America. Faus and Sujansky (2008) provide a comprehensive review of such systems. Although open-source development relies on voluntary participation of programmers and developers, several researchers looked into the economical plausibility of the open-source software model and found that the model relies on sound economic principles despite the initial impression that it offers something for nothing (Muffatto, 2006).

Several studies advocated the open-source model as a solution to the challenges that hamper the adoption of EMR software. One editorial note of the Journal of the American Medical Informatics Association (JAMIA) proposed that open-source software may overcome the traditional barriers of adopting EMRs, namely, excessive cost, the transience of vendors, and the lack of common data standards (Kantor et al., 2003). First, open-source software reduce ownership and maintenance cost by freeing the users from exclusive license and restriction on how to use and modify the software. Second, the user is free to change the vendor at any time; moreover, the disappearance of a vendor is not a threat to customer service because the support can be picked up by a different vendor. Third, open-source developers are known to embrace standards and encourage compatibility; therefore, the development of open-source EMRs will favor and push the usage of standard in the health IT domain. Similar results are reported in other studies (Safadi et al., 2010a).

Identification for an Opportunity for Contribution

The previous discussion outlines the potential of open-source EMR in the health IT domain. Yet, previous literature on this topic is scarce and focuses mainly on the adoption phase by considering the advantages of open-source EMRs vs. commercial counterparts.

To the best of our knowledge, there has not been any study that looked into the implementation process of an open-source EMR. Much of what we know about EMR implementation and usage comes from the experience of a few large hospitals affiliated to renowned medical schools which employ big commercial EMRs. More research is needed to understand how the implementation process of open source EMR unfolds and the relationship between technology design and its flexible adaptation to clinical workflows.

The research proposed here aims to perform a detailed field study of ambulatory clinics that are implementing open-source EMRs. Via interviews and participant observation in clinics, we expect to identify the unique dynamics of open-source EMRs. We focus our attention on the mutual effects between technology and practices and we hope that this study will enrich our understanding of the issues surrounding the transformation of medical work.

Conceptual Framework

As mentioned previously, EMRs require heavy customizations to comply with the legal requirements, drug nomenclatures and references, billing systems, integration with state agencies, and local practices. The flexibility of an EMR is crucial to achieve the required customizations. Moreover, customizations indicate a match between the EMR feature and the work practices in the medical setting. The absence of such customization may hinder the acceptance and the efficiency of the EMR.

Workarounds are a known but understudied phenomenon in Information Systems (IS). Workarounds appear when users do not comply with the intended and prescribed use of the system after implementation. Instead, they accomplish their tasks with unintended patterns of usage (Azad et al., 2008). In some instances workarounds may try to bypass the whole system at once (Koopman et al., 2003).

The two sides (workarounds and customization) could be seen theoretically to correspond to a structurationist approach. Structuration-based theories in IS such as adaptive structuration theory (DeSanctis et al., 1994) and situated change (Orlikowski, 1996) emphasize the mutual effects between people and technology. While these theories offer a broad description of the phenomenon, workarounds

and customizations are specific materializations of the structuration concept. This is because studying the phenomenon from a customization-workaround perspective implies paying attention at the details of people-system interaction. When people manage to change the system customizations occur and when people change their behavior to cope with the system, workarounds occurs.

Previous research on workarounds focused on the relationship between workarounds and previously established frameworks in IS such as resistance (Ferneley et al., 2006), and situated practices (Azad et al., 2008; Faraj et al., 2006). In addition, it has been argued that workarounds are undesirable because they reduce the efficiency of the system, and therefore increase the cost of medication (Azad et al., 2008; Koppel et al., 2008).

In the context of EMR implementation, workarounds are informal customizations of the system done by the medical users. Workarounds encode rich knowledge about both the system and the medical practices and about the missing and desired features of the EMR. Workarounds are more pronounced in open-source software because of the initial immaturity of the system and its flexibility and openness. In this sense workarounds present a risk and opportunity. They imply the medical users are unhappy with some aspects of the system, at the same time they encode rich knowledge about these worked-around desired aspects. If this knowledge can be transformed to formal customizations, both the quality of the EMR and the satisfaction of its users increase. Safadi et al. (2010b) proposed the knowledge embedded in workarounds transform to formal knowledge via a four-phase process that is similar to Nonaka's spiral of knowledge (Nonaka et al., 1995).

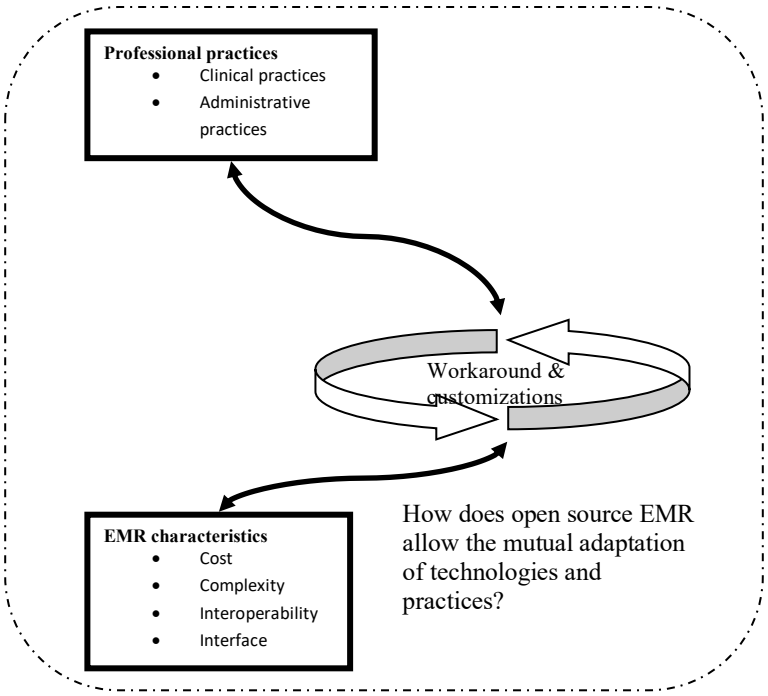
Error! Reference source not found. depicts the conceptual framework used in this exploratory study. We used the concepts listed here to help us focus our attention while conducting the study. At the center of the conceptual framework are the concepts of workarounds and customizations. As mentioned previously, EMR systems need to be tailored to the needs of the local site. This is formally done via customizations.

Workarounds and customizations represent the interaction between users and the system. The feature of the EMR and the behavior of the medical practitioners will determine the extent of customizations and workarounds that happen during the implementation. We focus on both the medical practices at the implementation site and the software characteristics of the open-source EMR. The match or mismatch between the two concepts—people and software—is captured by the concepts of workarounds and customizations.

Studied medical practices include clinical, front-desk, billing, scheduling and clinical administration. The scope, efficiency and complexity of these practices will be studied. Also, of a particular interest is the change in roles in the clinic if such a change happens during the implementation of the EMR.

EMR features include functional features such as clinical, business and administrative functions and non-functional features such as cost, flexibility and performance. In addition, open-source specific characteristics such as collaborative development, community sharing and support will be considered.

Figure 1: Conceptual Framework



Research Strategy

The data collected will include archival data, interview data and field notes gathered from on-site observations. Archival data to be collected include: project reports, systems requirements documents, user manuals, internal meeting minutes, presentations, project reports, system documentations, request for change, and process descriptions. Interviews will be used to externalize mental models about the artifact, its features, and the users. Observation will enrich our understanding by providing access to dynamics of interaction. The proposed study will be longitudinal in nature and the research method s will include:

- Participant observation (e.g. at user group meetings, or development conferences);
- Interviews of key informants;
- Site visits to the clinics which will be adopting or have adopted and implemented an open-source EMR in Canada;
- User group mailing list observation and discourse analysis for each of the user groups and for the developer community.
- Surveys may also be used in later stages of the study.

This approach will provide sufficient triangulation in methods, sources, informants, and research sites to ensure valid findings. The proposed research will be conducted over a period of 3 years, however, for the purpose of this pilot study; data will be limited to interviews and on site observations. In addition, only one site that is implementing an open-source EMR is studied. The studied site was selected because it is affiliated with our research institution. Future sites will be selected according to the geographic area (i.e. province) and size. The unit of analysis is the individual medical worker.

The Site

We are studying the implementation of an open-source EMR system at a medium-size walk-in clinic in a major Canadian city. The clinic is affiliated with the medical center of university and serves approximately 30,000 patients per year. Its staff complement includes 38 physicians, 15 nurses, and 10 clerks. The clinic switched from paper charts to EMR last year. They have been using paper charts for a long time, but decided to switch to an EMR to reap its benefit. The switch is gradual and the two systems are still operating together with some physicians still using the paper charts.

The Technology

Open Source Clinical Application Resource (OSCAR) is an open-source EMR developed at the Department of Family Medicine at McMaster University and deployed in multiple clinics in Canada including clinics in British Columbia, Ontario, and Quebec. OSCAR is indeed a suite of medical applications that integrate with each other. The OSCAR system offers a set of integrated applications, including: OSCAR EMR, the focus of this paper; MyOSCAR, a patient controlled personal health record; MyDrugRef, a collaborative resource for prescription medication; OSCAR CAISI; and OSCAR Resource.

The default installation of OSCAR EMR runs on a stack of open-source applications. Linux is the default operating system, although it is possible to install OSCAR on Windows. OSCAR uses two databases: MySQL for patient data and PostgreSQL for drug data (interfacing with DrugRef, a Canadian standard database of drugs). In addition, Apache is the default web server. OSCAR itself is written in Java Enterprise Edition with HTML and JavaScript code which makes it highly portable to different platforms.

OSCAR EMR features a web-based interface that allows multiple users to interface with the system simultaneously through different browsers and platforms. The web interface also allows accessing the system over the Internet through secure connections which is very handy for geographically dispersed organizations. OSCAR also features multiple administrative and clinical functions. Administrative functions include demographic management, appointment scheduling, reports and billing. Clinical function includes electronic charts, drugs and prescriptions, consultation, flow sheets, laboratory and many other features. Besides built-in function, OSCAR allows the rapid development of plug-in medical forms. This features helps users customize the system without much technical knowledge, all what is required is a basic knowledge of HTML. The studied clinic exploited this feature in order to customize many aspects of the system as discussed in the next sections.

In the studied clinic, the decision to adopt the open-source system (OSCAR) stemmed from its reduced initial and maintenance costs. When evaluating the different alternatives, the clinic found that the open-source EMR system will cost fifth of the cost of alternative commercial solutions. In addition, developed by a Canadian university, OSCAR tailors to the particularities of the Canadian healthcare sector.

The Methodology

We adopted an observational methodology in this study combined with key informant interviews. We also interviewed different stakeholders and key informants from other clinics that were involved in the implementation. In addition to observations and interviews, the first author took an active role in the implementation process and helped in making modifications and customizations to the system. This helped him understand the workflow and the EMR system. It also allowed him to gain legitimate access to the clinic and facilitated access to resources and people for the purpose of the research.

Data Collection and Analysis

We started attending weekly meetings at the clinics since July 2008 when the system was installed. We attended training sessions and observed meetings related to the system. We did not have a preferred theoretical lens when we were first engaged in the study; the theoretical perspective emerged during the first year of observations. The meetings were taped and field notes were taken.

Four stakeholders were interviewed during the course of this study: the clinic manager who is also a physician, the clinic main secretary, a doctor, and a nurse. Different roles were chosen in order to capture the different perspectives among medical workers in the clinic. The interview followed a semi-structured setting were a predefined set of questions were prepared, however the interviewer allowed the interviewee to develop on his/her answers and followed up with additional questions that are not in the guidelines. The interviews were taped and transcribed afterwards. They were then coded and analyzed using AtlasTi Software.

In this exploratory study we are adopting a narrative strategy to analyze the data. The narrative strategy allows sense making and grounded theory construction and it is suitable for research with ambiguous boundaries (Langley, 1999). The narrative is constructed from both the observations in the clinic and the follow up interviews with the key informants.

Data Collection and Analysis

The narrative is organized chronologically by the different phases of the EMR implementation in the clinic. In each phase, we describe the behavior of the medical workers in the clinic—i.ie people—the EMR—i.e. software. Since we are using the theoretical lens of workarounds and customizations to capture the interaction between the users and the system, the impact on practice and the state and also the evolution of workarounds and customizations is described for every phase.

Phase I: decision making. The decision to adopt an EMR in the clinic stemmed from the clinic manager who worked in Europe before joining the clinic in Canada. The manager was surprised by the lack of an electronic system and decided to acquire one.

"*Yes, so I was shocked to find there wasn't one … The project was necessary because we have 45,000 patients and we have no room for paper charts, there is no conceivable way we can manage the clinic without a system" (Manager)*

The acquisition phase consisted of comparing several commercial EMRs and OSCAR the open source EMR. The decision to adopt OSCAR was mainly due to its reduced cost and its support to academic mentorship:

"Yes, so for the first 5 years I have looked at OmniMED very seriously, I looked at, before we implement it, I looked at EMIS very positively … I looked seriously at 5 separate systems. 6th is OSCAR. … So yes, that is 7 systems I considered and the main barrier for all of them was price … and its (OSCAR's) ability to work in academic setting … You need to come and sign the chart, for any entry done by a student or a resident you need to be able to come and sign it electronically in such a way that your changes to it are the ones that appear on the screen the first and only afterwards can someone come and see what you changed." (Manager)

From the clinic perspective, there was doubt about the system and resistance toward it. This resistance was absorbed by the persistence of the manager to put the system in place. Therefore, everyone was expecting the system soon so when it was finally implemented there was no overt resistance to it.

"Yes, so I started talking about EMR the day I arrived here. So people were shocked. No sorry I was shocked and people started getting the idea that I was saying, you know, this is a developing country; this is not a developed world system. And, so they had 5 years of me going over them and say: why don't you get an electronic record … That was three years ago, I said: OK, well I will just go and do it myself and just positive deceitfulness I will go behind that back and put an end then ask for permission. But the point about is, about me talking about it for 5 years is that everyone, absolutely everyone in the department every member staff know I was totally committed to this and this is what we should do and this is been dripping for 5 years. So when it actually arrived, it wasn't a surprise." (Manager)

There were *some* concerns about the software, in particular maturity and security, those concerns were eased up by the persistence of the manager to get the EMR:

"OSCAR just reached the stage where it was implementable … it is very secure, everything is encrypted and transmission between the server and the machine that you are looking at are encrypted. It has gone through Ontario accreditation repeatedly the third iteration in that accreditation, so there wasn't really any concern from that point" (Manager)

It is also worth *mentioning* that there were no concerns regarding the open-source nature of the EMR because of the organization culture. In particular the initial zero cost of OSCAR did not cast doubt on its effectiveness.

"They recognized that it is a little bit clutchy in certain places because they don't have to pay any money for it they accepted more that clutichiness … because in primary care medicine a lot of what we do we share free. I mean every single publication I do is shared free, people are making profit from it but it

is on Medline available to anyone. So there is a source of sharing anyway, and when you tell that to people, yes, we share everything we do, we don't sell our knowledge, we tend to share it" (Manager)

Phase II: training and initial experience. The second phase is characterized by the first interaction with the EMR in the clinic. The EMR was received differently by various practitioners. The staff followed a formal training period provided by the software developers.

"We had Dr. C coming and giving a presentation for us. So J was also a key factor for us on the programming side *to help us learn the program itself. We got half a day of training, so a lot of it as you know was touch and go, we would try things that didn't work, we try something else, and if we are really stuck, then we recall our handy." (Secretary)*

After the training, some physicians got excited and became champions of the EMR. Their role will be important in promoting the EMR and supporting it among colleagues.

"In the beginning it is of course normal that everyone does these additional tasks. We have some; we like to call them, superusers. Some of them are doctors who are very excited and have become completely implemented in helping others in the group going into the system." (Secretary)

"Once I saw it has been a fit I promoted it to my colleagues and one of the promotion activities was at the Monday's luncheons where we have invited other doctors to use it with success, and we also visited other sites." (Physician)

"There were doctors who said: "It got to be used by everyone" and they forced this issue and they became patient with everyone using it, so they did that. I did not tell them to do it, they would have to log on and had to read someone writing and could not read it and got more frustrated, and so they said: "you have got to use it"." (Manager)

Other physicians *just* got along with it and some refused to use it due to lack of computer skills. However, it is important to mention that the open-source nature of the EMR reduced the tension of having to use the system between the management and the physician in the clinic. Since the EMR is free the management did not feel pressed to oblige physicians to use it.

"There were one or two laggards that are still resistant to it. There are keyboard skills, people not familiar with. But because the system is free, you don't have that enormous pressure to switch on everyone immediately" (Manager)

"I think one of the negatives of the private systems is that first of all, it is expensive and because of that there is a great pressure to use it, so you cannot deploy it slowly in a comfortable way with all adopters." (Physician)

The EMR required a lot of customizations in the beginning. Fortunately, the OSCAR EMR was flexible enough and *allowed* various customizations to be done locally. The flexibility was well received in the clinic as an indication of the quality of the EMR and the open-source software in general:

"That was, not really a surprise, but we had to customize the clinic setup, we had to put in all the doctor names, nurses names, give them roles within the system which give some level of access, we had to set up the address templates so when prescription are printed we will have them correct … And then the clinical customizations, the templates, that is just an ongoing process and it continues where we continue to put it into place." (Manager)

"The beauty of the open-source, is that we can go in and setup the system the way we want it to run for each individual clinic and being such a diverse group with different types of clinics, that is the beauty of having open-source and being able to do conveniently change no matter what it is: eforms, the way our notes are being put by the doctors, the way that notes are being put in, we can setup templates which is quite helpful as well." (Secretary)

"Certainly customizations for forms and templates were one of the first things that we realized, but then we realized how easy it was. To put a template it took me I think about 10 minutes. And I did three of them in one hour. So that was a big surprise" (Physician)

There were some ongoing concerns about security and support. Given that the EMR is free, one should not expect good support to it:

"Patient confidentiality it is always a concern for us. But when the system was implemented that fear dissipated because *even if it is open-source, the open-source aspect of the EMR is actually for information purposes, whereas each site that has implemented that particular system that is called OSCAR has its own server. SO it is a very secure system." (Secretary)*

"There were two concerns that I had. One is security and one was support. Because when you are paying for something, it may be expensive but usually you get a response" (Physician)

This *fear* disseminated quickly when the clinic realized that OSCAR's open-source community provides extensive help and support via the online group. Moreover, this support is free and this helped the clinic to make extensive customizations to the EMR:

"The user group was pretty useful and we still use that but in the first month certainly we were asking the user group a lot of questions like any new users does, and some of them were about customization." (Manager)

"The beauty of open-source it is when you ask one question and you get multiple answers to a question that we had. So we would try them and find the one answer that works. For instance, the toilet paper roll, as we call it, this is where the doctor writes his notes on the patient. We were not sure how to customize it, to go and add some *templates into the system, so one question and one answer later we were able to do that among our group." (Secretary)*

"And there used to be someone on-line who will get someone to do it and to put on a text bar into something *will cost $500 and take a week and famous line of all that is "this is a big project" when indeed it is only a text box ... With OSCAR you got a listserv with all questions answered already. So if you are professional with listserv you can send a question and get it answered." (Physician)*

During the initial period we observed the emergence of many workarounds. These workarounds were overt and were not recognized by the users as workarounds. Failure to customize the EMR or the lack of a certain feature pushes the users to develop workarounds. We will give an example of one workaround that became a standard practice in the clinic: searching for a drug.

OSCAR integrates with DrugRef, a Canadian database of drug names and interactions. The database is extensive and OSCAR provides a search interface that helps searching for a drug based on its name. In addition, once the physician prescribes the drug, he can add it to a list of favorite drugs that shows up in the left pane of the E-Chart screen. In one training session, the presenter was explaining the drug-related functionalities of OSCAR and demonstrating the search functionality. Attending physicians pointed out that the database contains a lot of drugs that they are not familiar with. One physician noted: "I am not familiar with all of these drugs. They remind me of the old days of med school. We certainly need a subset of them."

In a later training session, some physicians started complaining about the drug interface. The drug database does not include commercial names of drugs. Also, commercial brands differ between Canada and the US. The drug database is extensive and upon issuing a query, the search function returns a large list of drugs based on similarity of the name. For example, a search for aspirin may return more than twenty aspirin drugs differing in the form and the dosage. Assigning each prescribed drug as a favorite quickly fills the favorite list. In addition, the favorite list does not allow classifying the drugs or organizing them in a hierarchical fashion. One physician pointed out that he was assigning his favorite drugs to the "test patient", a dummy patient that comes with the initial installation of OSCAR. Assigning the drugs to the test patient allows the doctor to consult them later to copy and paste the drug name.

Assigning drugs to the test patient proved popular. In order to classify drugs several dummy patient records were created. The "Cardio Patient" is a dummy patient to whom popular cardiovascular diseases are assigned. "Rheumatoid Patient" is another dummy patient to whom arthritis drugs are assigned. The usage of dummy patients spread among many doctors who refined the list of the drugs assigned to those patients and used them and favored them over the default search interface for common drugs.

Phase III: ongoing usage. With the continuing usage of the EMR in the clinic, medical practitioners realized its benefits. These benefits are also changes *to* the standard work practices in the clinic.

"What you are doing is running your clinic very effectively and after a year people realized that. So there are people *who are still fighting it a little bit but you can't actually look after the patient in the clinic without using it, because all the triage is done by the nurses via the electronic record, so they can't work in the clinic without it." (Manager)*

"The level of patient care has probably increased by fold … the level of care I think has gone up quite a bit since we had the system implemented … What the beauty of this one, this EMR, that is implemented in our clinic, is that you see the doctors' notes, you see all of the consultations, you see all of the results on one screen, on one menu. So you don't have to go searching for patient information, and this is critical when you have a patient coming in with trauma or needs urging care." (Secretary)

"It is much more efficient as opposed to getting a paper chart and trying to find a particular data that you are looking for, it is right there so it is efficient … it is much more effective because the information is all right there as opposed of doing a head to toe questionnaire with the patient again for the twelfth time, so it is very efficient" (Nurse)

The physician we interviews indicated that the EMR helped him to coordinate his work at different sites. The ability to access *patients'* charts remotely is a great advantage for physicians who are affiliated with different clinics:

"I become more efficient, I am not searching through documents to find a document in my chart. I know if I have my chart scanned in or not. I can treat patients at distance, like for home care or when people call me at home for something. I can access charts of the patients who have been to the urgent care clinic and then come to my office and vice versa I can see the patients' charts in the walk-in clinic that I have seen them there. I can access blood results and radiology results within seconds." (Physician)

The EMR also altered the communication patterns in the clinic. The messaging system allows for more efficient *communication* among colleagues:

"Coordination improved it because you could read what other people have written which is enormously helpful as the system became more and more used you start read what people prescribe and then you can re prescribe drugs, you can see the allergies, so eventually the electronic record started to help care." (Manager)

"With the messaging system and the tickler system, communication with my secretaries and nurses is much improved and my colleagues. For example, I did see a patient in the walk-in clinic, it was a sore throat that turned out to be mono necrosis that was missed by a previous doctors. Once the test was come positive I made a comment on the chart and sent a copy to the doctor involved so she is aware of that diagnosis. But I get 4-5messages a day about blood tests, about patients in trouble. It is much, I mean, I probably won't be getting 50% of these messages in my day to day work as I have to sit down and talk to the people." (Physician)

The gained work efficiency was boosted by the flexibility of the EMR and the added customizations. *Customized* templates for consultations and referrals improved the external communication with clinics and specialists with resulted in a better consultation and referral practice:

"The referral to our colleagues is much improved because we automatically include a list of medications, we can have a past history and we can copy and paste these templates, even our notes into the referral. And printing out the old results is a breeze, you don't have to go and ask your secretary for a copy. Just click it open and print it. I am sure our specialists colleagues are very pleased with OSCAR, as are the pharmacists who can read our perceptions." (Physician)

It is clear that the *customizability* of the EMR was essential for its acceptance in the clinic. Customized forms matched clinic paper forms and customized templates helped to match work practices. Moreover, the software extensibility facilitated complying with government regulations:

*"The e-*forms *are very essential, some of the are governmental, so they facilitate the task for us of having to write the information on paper forms and then scan them into the patient charts and then send them off." (Secretary)*

At this stage, medical practitioners kept using the workarounds mentioned in the previous stage. However, they became aware of other workarounds—pronounced workarounds. Realizing that the EMR does not integrate with other electronic systems, the staff at the clinic set up workarounds to deliver data

from these systems to the EMR. A notable example is integrating lab results to patients' records. This is done by getting the lab results via electronic fax link to the laboratory and then adding the digital version to patients' records. This workaround was recognized by all of the interviewed informants and was described as an inconvenient by necessary step toward the full integration:

"What is happening now is that we created a shortcut on the existing desktops, which mean the secretaries, the front-end and the nurses desktops, via fax, so the faxes are coming in and being dropped into a folder on the desktop which is been viewed by the nurse." (Secretary)

"Well, we have just sent the hospital money to connect us to download of our blood tests to our charts, and that I believe is a standard practice, I don't know if it is a workaround but that is something we had to do, to pay the hospital to do that. I mean the problem is that our radiology departments don't send out HL7 reports, they send out faxes. They are upgrading their computer systems, so hopefully that will be improved." (Physician)

Synthesis

Medical practitioners in the clinic were unaware of the importance of having an EMR until the new manager joined the clinic and took the decision to acquire one. The decision making period took a while, however, the manager made sure that the best system is selected. Interaction between the users and the EMR started after the initial implementation and the training period. During this stage medical workers realized the extent of customizations required to make the system comfortable for them to use. Few doctors refused to use it while some became champions and promoted the system inside the clinic. The champions along with the administrative staff were responsible of most of the customizations to the system. These customizations were done locally with a great help from the EMR online community. The open-source values of freedom and sharing complied well with the medical tradition in family medicine which helped to increase the system acceptance in the clinic.

"In primary care medicine a lot of what we do we share free. I mean every single publication I do is shared free, people are making profit from it but it is on Medline available to anyone. So there is a source of sharing anyway, and when you tell that to people, yes, we share everything we do, we don't sell our knowledge, we tend to share it." (Manager)

Moreover, the management did not feel compelled to push the system to everyone because the system was relatively cheap to acquire. Not all wish features were realized though. Therefore workarounds emerged and diffused in this stage.

With the ongoing usage of the EMR, its benefits became clearer and outnumbered the initial annoyances. The EMR increased the efficiency of communication and coordination in the clinic. The web interface greatly helped physicians who work at different sites and who need to access patient data remotely. The new customizations helped to achieve better communication with external parties such as specialists. At this stage some workarounds remained became a standardized practice of the system usage. However, users became aware of other workarounds as they performed them for doing tasks not supported by the system.

Our findings around the importance of workarounds in the implementation of an EMR have implications for better understanding the implementation of IS systems in health care. Given the complexity of medical work, the high reliability aspect of medical work, and the importance of coordination practices in health care, IS researchers need to take the health care context seriously (Chiasson et al., 2004; Davidson et al., 2005; Faraj et al., 2006). Not only are workarounds common and usual during the implementation of commercial EMR (Azad et al., 2008), but the issue becomes increasingly salient during the implementation of an open source EMR. Given the heavy reliance of the open source development model on feedback and input from users, user initiated change and workarounds are likely to become a normal and essential part of the IS implementation process. The implication of such an emergent phenomenon, as shown in the preliminary results of our research, would be to make workarounds an essential aspect of health IT implementation.

Implication for Practice

DesRoches et al. (2008) report that short term cost of acquiring EMRs is a major barrier for their adoption in the US. Commercial vendors use the "software as a good" business model. On the other hand, open-source software relies on the "software as a service" business model. Open-source EMRs subsidize the initial big acquisition cost over yearly maintenance and support. Moreover, the client is free in choosing his support providers if any. It should be noted, however, that this freedom comes with a cost—it requires commitment and dedication from both the developers and the users to the system.

Open-source software development relies on pooling distributed talent and skill and organizing it over the Internet. One of the challenges of open-source projects is to keep the development team motivated and enthusiastic. Projects that gather traction will attract more talented and dedicated developers. This creates a snow-ball effect where active projects will draw more users and so on.

The studied clinic chose and open-source EMR because of its cost advantage over commercial competitors. However, it should be noted that cost was not the only factor. Several EMRs were looked at and compared before picking OSCAR the open-source EMR. Indeed, the clinic manager ensured that OSACR was matured and well suited to the clinic needs. In addition, he made sure that the development community is active and enthusiastic about improving and supporting the EMR.

The clinic manager was dedicated to the project from the beginning. In addition, early champions helped in promoting the system and customizing it for the rest of the clinic. In doing so, they engaged in a fruitful communication with the development and the user community over the Internet. The online community provided answers, support and customizations as well. This helped them to accelerate their learning and increase their knowledge about the system.

To conclude, open-source software presents a unique opportunity to obtain a low-cost highly-customized EMR. However, the implementation process is largely different than that of proprietary EMRs where a single party implements, supports and maintains the system. The implementation of an open-source EMR requires the dedication and participation of all stakeholders in the clinic. It also requires active communication and interaction with the developer and user community. Notwithstanding these requirements, the overall result is a much less expensive system, more customized to the local clinic practices with knowledgeable and well informed users.

Implication for Future Research

The implementation of electronic medical record systems is a complex process that encompasses many technical and social issues. This preliminary study represents a research opportunity by focusing on the impact of open-source EMRs on clinical practice via the theoretical lens of workarounds and customizations. This study has taken a narrative approach to track people's behavior toward the EMR during the implementation process. The study focused on one clinic that is implementing an open-source EMR.

In the conceptual domain, future work aims to develop a practice based model of technology evolution that emphasizes the role of leader technology framing and social movement concepts to explain how open-source EMR development takes place and how it affects clinical practice. In order to achieve this goal, the theoretical framework should be extended to become more grounded in the observations with the goal of producing a complete and validated theory at the end of the research project. This theoretical advancement requires further work in methodology and data analysis.

In the methodological domain, future work includes studying other sites that are implementing and using open-source EMRs and collecting different types of data such as archival data, project documents, and online community mail list. Other technique may be used to enhance existing data. For example, in order to better capture practice, an interview to the double technique may be used (Nicolini, 2009). In addition, more analysis is required, for example discourse analysis for each of the user groups and for the developer community. Surveys may also be used in later stages of the study. This approach will provide sufficient triangulation in methods, sources, informants, and research sites to ensure valid findings.

Limitations

Notwithstanding the general limitations such as limited data, one case design and the selection of key informants, we focus our attention here to limitations specific to the study. More precisely we discuss two issues: the differentiation between open and close source EMRs, and the theoretical perspective taken to study the implementation.

The premise of this study is that open-source EMR will differ than proprietary competitors in many aspects. However, the distinction between open-source and proprietary software blurs in many domains. Although open-source software is associated with more flexibility and freedom to change and modify the software, many proprietary systems provide extensibility features and allows for a high degree of customization. Indeed, many commercial vendors try to include distinctive features of open-source competitors such as customizability and community support in their products. For example, Microsoft Office has built-in programming support to allow for new functions and capabilities. In addition, Microsoft hosts a huge online community for office users. Our limited knowledge of commercial EMRs inhibits us from making a fair comparison with OSCAR the open-source EMR. However, from our interviews it seems that in the medical setting, commercial EMRs are still associated with close development and limited functionalities.

"One of the negatives of the private systems ... when you want something done to change, it is also extremely expensive ... the third aspect is anyway in private systems is one you come to renew your contract after three years, they usually sign you on a cheap deal. And when you renew it, suddenly it is three times the price, and you have all your data on it and they captured you, and you cannot transfer your data so you are locked in." (Physician)

More research is required to answer this question. In particular we are looking to gaining access to a site deploying a proprietary EMR to make an informed comparison.

The implementation of electronic medical records is a complex process that envelops many technical and social issues. To focus our research an observation we have used the framework of workarounds and customizations. This framework emerged from our initial observations and it helped us to concentrate on specific concepts while doing field observations and conducting interview. However, it may be limiting to discard other established theoretical lenses such as resistance (Ferneley et al., 2006), and situated practices (Azad et al., 2008; Faraj et al., 2006). These theoretical perspectives shed light on other aspects of the research problem and it may be helpful to consider them in future theoretical development.

Conclusion

Electronic Medical Record (EMR) systems are infamous for their complexity and inability of to comply with the long established medical practices. In this paper, we suggested that workarounds and customizations may be suitable to model users' behaviors toward the EMR. Customizations and workarounds encode sophisticated knowledge about the system. Users' formal training, experience, and social interactions are the ingredient of customizations workarounds. We have validated our conceptual framework with data gathered during one year of observations and interviews of a medium size clinic implementing an open-source EMR system. Our preliminary results indicate the potential of using workarounds and customization as a theoretical lens to study users' behavior in an open-source development setting.

This research is still in progress, we have not yet presented a cohesive theory that describes and explains the dynamics of open-source EMR implementation. We propose that workarounds will be more pronounced in open-source software because of the flexibility and active participation of users in the system development. We need to expand the study to encompass more sites, in depth observations, and to collect different types of data.

References

Azad, B., and King, N. "Enacting computer workaround practices within a medication dispensing system," *European Journal of Information Systems* (17:3) 2008, pp 264-278.

Baron, R.J., Fabens, E.L., Schiffman, M., and Wolf, E. "Electronic health records: just around the corner? Or over the cliff?," *Annals of Internal Medicine* (143:3) 2005, p 222.

Beaudry, A., and Pinsonneault, A. "Understanding User Responses to Information Technology: A Coping Model of User Adaptation," *MIS quarterly* (29:3) 2005, pp 493-524.

Berg, M., and Bowker, G. "The multiple bodies of the medical record: toward a sociology of an artifact," *The Sociological Quarterly* (38:3) 1997, pp 513-537.

Chiasson, M., and Davidson, E. "Pushing the contextual envelope: developing and diffusing IS theory for health information systems research," *Information and Organization* (14:3) 2004, pp 155-188.

Davidson, E., and Chiasson, M. "Contextual influences on technology use mediation: a comparative analysis of electronic medical record systems," *European Journal of Information Systems* (14:1) 2005, pp 6-18.

DeSanctis, G., and Poole, M.S. "Capturing the complexity in advanced technology use: Adaptive structuration theory," *Organization science* (5:2) 1994, pp 121-147.

DesRoches, C., Campbell, E., Rao, S., Donelan, K., Ferris, T., Jha, A., Kaushal, R., Levy, D., Rosenbaum, S., and Shields, A. "Electronic Health Record Adoption in the Ambulatory Setting: Findings from a National Survey of Physicians," *N Engl J Med* (359) 2008, pp 50-60.

Faraj, S., and Xiao, Y. "Coordination in fast-response organizations," *Management Science* (52:8) 2006, p 1155.

Faus, S., and Sujansky, W. "Open-Source EHR Systems for Ambulatory Care: A Market Assessment," California Healthcare Foundation.

Ferneley, E.H., and Sobreperez, P. "Resist, comply or workaround? An examination of different facets of user engagement with information systems," *European Journal of Information Systems* (15:4) 2006, pp 345-356.

Garrido, T., Jamieson, L., Zhou, Y., Wiesenthal, A., and Liang, L. "Effect of electronic health records in ambulatory care: retrospective, serial, cross sectional study," *British Medical Journal* (330:7491) 2005, p 581.

Hillestad, R., Bigelow, J., Bower, A., Girosi, F., Meili, R., Scoville, R., and Taylor, R. "Can electronic medical record systems transform health care? Potential health benefits, savings, and costs," *Health Affairs* (24:5) 2005, p 1103.

Jha, A.K., DesRoches, C.M., Campbell, E.G., Donelan, K., Rao, S.R., Ferris, T.G., Shields, A., Rosenbaum, S., and Blumenthal, D. "Use of electronic health records in US hospitals," *The New England journal of medicine* (360:16) 2009, p 1628.

Kaissi, A., Kralewski, J., Dowd, B., and Heaton, A. "The effect of the fit between organizational culture and structure on medication errors in medical group practices," *Health Care Management Review* (32:1) 2007, p 12.

Kantor, G.S., Wilson, W.D., and Midgley, A. "Open-source Software and the Primary Care EMR," *Journal of the American Medical Informatics Association* (10:), November 1, 2003; 2003, p 616.

Koopman, P., and Hoffman, R.R. "Work-arounds, make-work, and kludges," *IEEE Intelligent Systems*) 2003, pp 70-75.

Koppel, R., Wetterneck, T., Telles, J., and Karsh, B. "Workarounds to barcode medication administration systems: their occurrences, causes, and threats to patient safety," *Journal of the American Medical Informatics Association* (15:4) 2008, p 408.

Langley, A. "Strategies for theorizing from process data," *The Academy of Management Review* (24:4) 1999, pp 691-710.

Lapointe, L., and Rivard, S. "A multilevel model of resistance to information technology implementation," *Management Information Systems Quarterly* (29:3) 2005, p 6.

Martin, J. *Cultures in organizations: Three perspectives* Oxford University Press New York, 1992.

Muffatto, M. "Open source: a multidisciplinary approach," Imperial College Pr, 2006, pp. 71-90.

Nicolini, D. "Articulating Practice through the Interview to the Double," *Management Learning* (40:2) 2009, p 195.

Nonaka, I., and Takeuchi, H. *The knowledge-creating company: How Japanese companies create the dynamics of innovation* Oxford University Press, USA, 1995.

Orlikowski, W.J. "Improvising organizational transformation over time: A situated change perspective," *Information systems research* (7) 1996, pp 63-92.

Rivard, S., Lapointe, L., and Kappos, A. "A Culture-Based Theory of Clinical Informaiton Systems Implementation in Hospitals," *mimeo HEC Montreal*) 2009.

Safadi, H., and Faraj, S. "Impact of Open-Source Software in Adopting and Implementing Electronic Medical Record Systems," Conference of the Administrative Sciences Association of Canada, Regina, SK, Canada, 2010a.

Safadi, H., and Faraj, S. "The Role of Workarounds during an Open-Source Electronic Medical Record System Implementation," International Conference on Information Systems, Saint Louis, Missouri, 2010b.

Sommerville, I. *Software Engineering 6th Edition* Harlow, UK: Addison-Wesley, 2000.

Tamblyn, R., Huang, A., Kawasumi, Y., Bartlett, G., Grad, R., Jacques, A., Dawes, M., Abrahamowicz, M., Perreault, R., Taylor, L., and others "The development and evaluation of an integrated electronic prescribing and drug management system for primary care," *Journal of the American Medical Informatics Association* (13:2) 2006, pp 148-159.

Ventres, W., Kooienga, S., Vuckovic, N., Marlin, R., Nygren, P., and Stewart, V. "Physicians, patients, and the electronic health record: an ethnographic analysis," *The Annals of Family Medicine* (4:2) 2006, p 124.

Walker, J., Pan, E., Johnston, D., Adler-Milstein, J., Bates, D.W., and Middleton, B. "The value of health care information exchange and interoperability," *Health Affairs* (19:01) 2005.